

Lec_17

Hardest Questions of GATE

part 1:

0-16

0, θ , $\sim$, $\circ$, ω

comparision of

functi

①

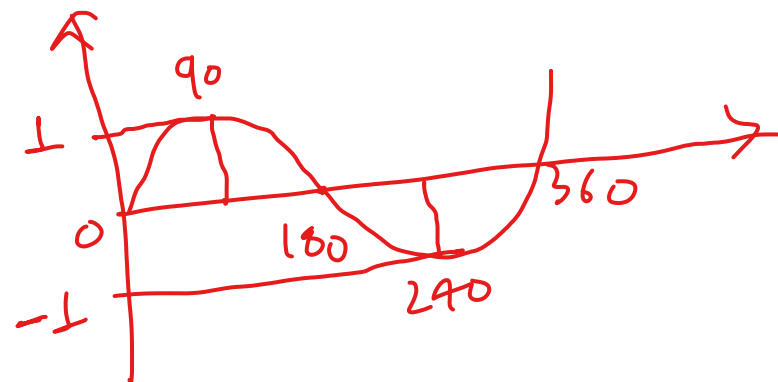
Compare two fn (given)

1.) $f(n) = n^{\sin n}$

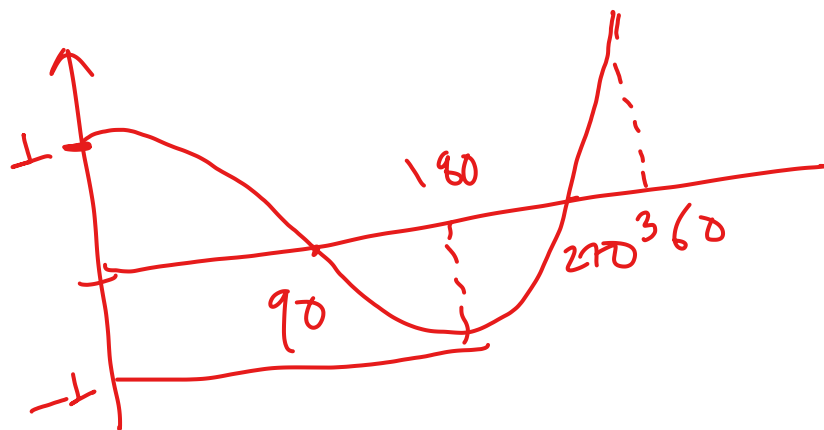
2.) $g(n) = n^{2 + \cos n}$

Solve

$\sin n$ an graph: .



cos n

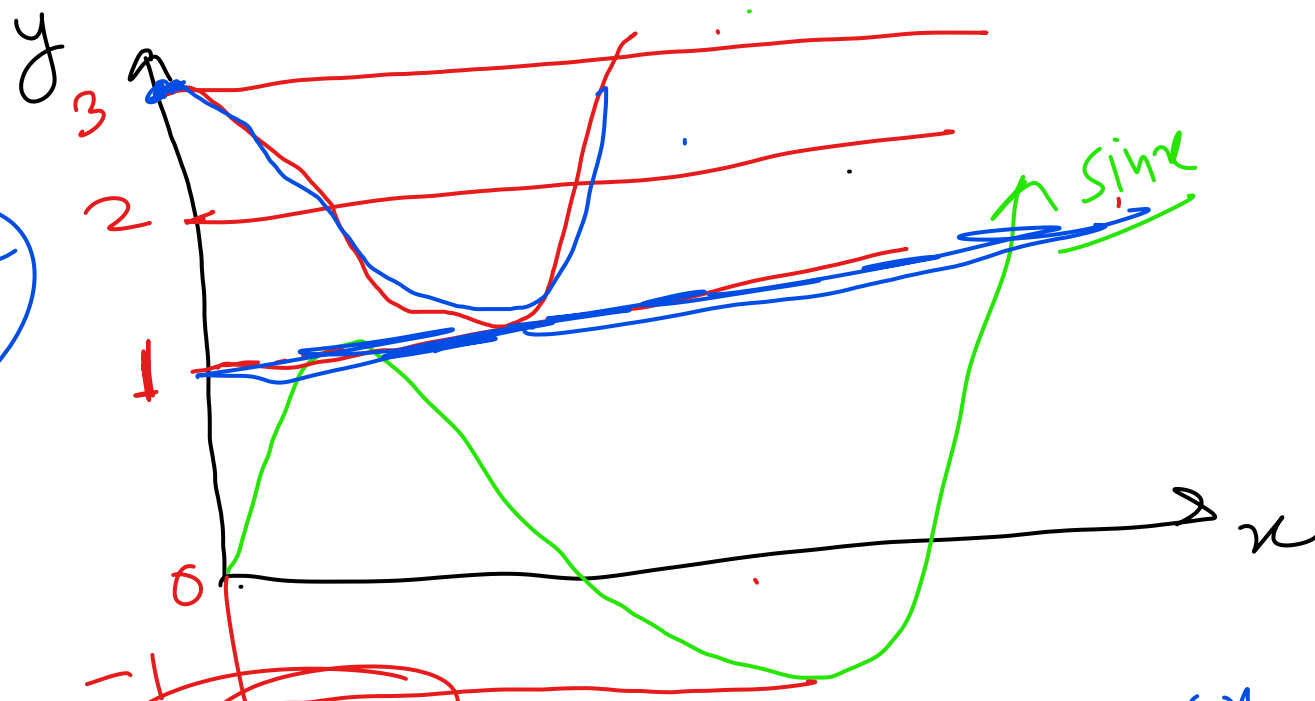


~~for~~

$f(n) = \underline{n}^{\underline{\sin n}}$

$g(n) = \underline{n}^{\underline{2 + \cos n}}$

$2 + \cos x$



$2 + \cos x$

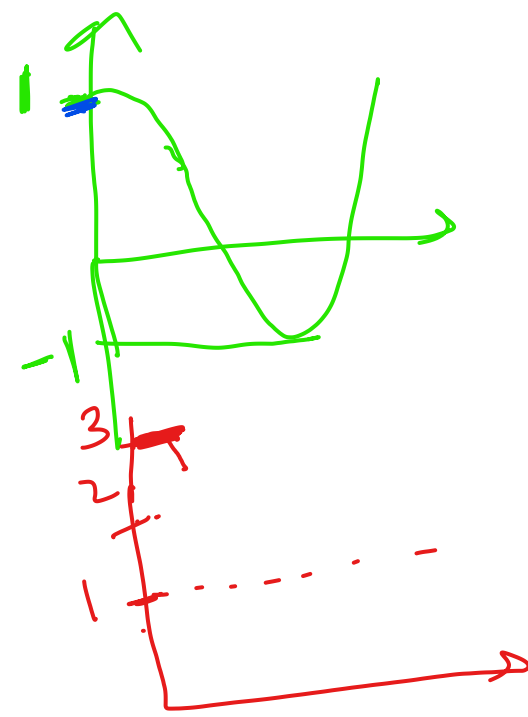
$\sin x$
 $2 \sin x$

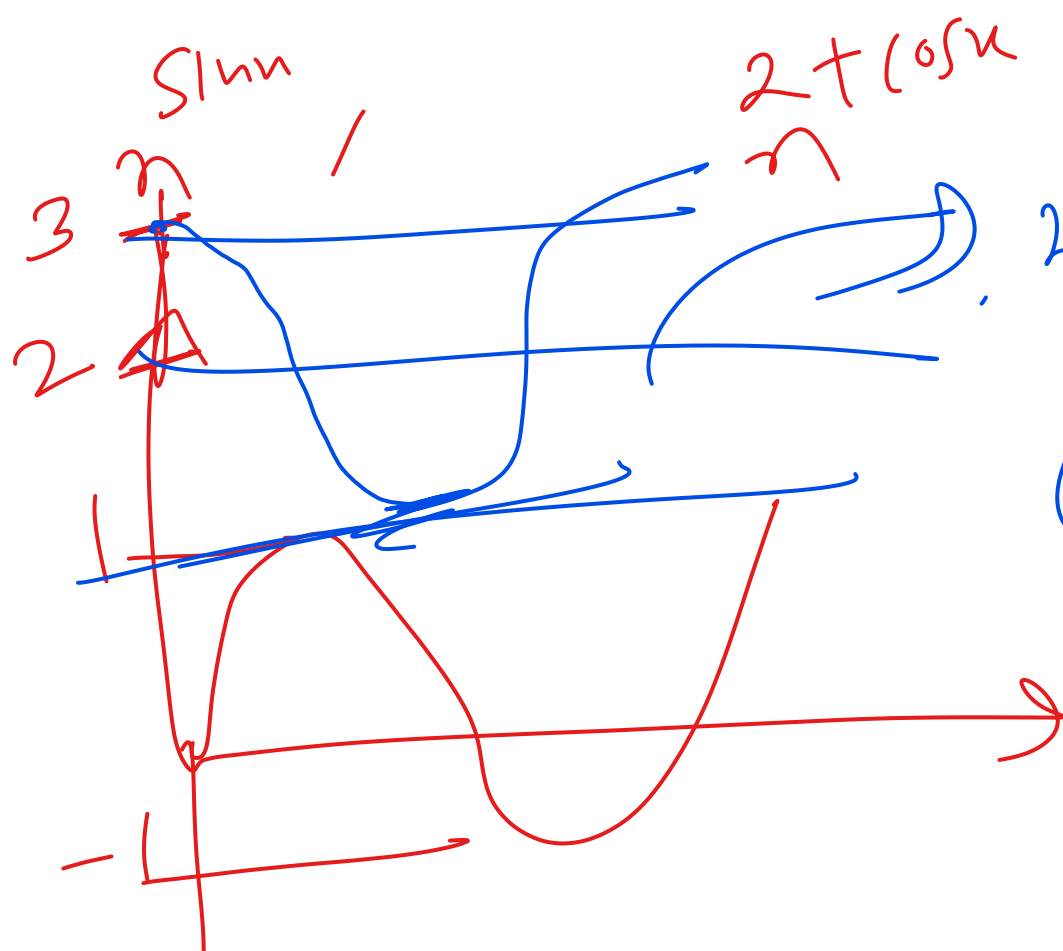
~~Handwritten scribbles in red and blue ink.~~

$2 + \cos x$

$2 + \cos x$

$2 + \cos x$
 $x=0$





$2 + \cos x$ is above

$$2 + \cos x \geq \sin x$$

$2 + \cos x$ is Bigger

②

$$f(n) = \begin{cases} n^3, & \text{if } 0 \leq n \leq 10000 \\ n^5, & \text{if } n \geq 10000 \end{cases} \Rightarrow \text{for small } n \rightarrow \text{no}$$

$$g(n) = \begin{cases} n^7, & \text{if } 0 < n < 100 \\ n^3, & \text{if } n \geq 100 \end{cases} \quad n=100$$

compare

$$f(n) = n^5,$$

$$g(n) = n^3$$

$$f(n) \geq (\cdot g(n))$$

$$n^5 \geq (\cdot n^3)$$

$$f(n) = \text{su } \underline{\underline{g(n)}}$$

3.

$$f(n) =$$

$$g(n) =$$

$$n^2 \times \log n$$

$$\cancel{n^2} \times \log n$$

$$\log n$$

$$\log \log n$$

$$\log n \times n^2$$

$$n^{2.01}$$

$$n^{2.01}$$

$$\cancel{n^2} \times n^{0.01}$$

$$n^{0.01}$$

$$\log n$$

$$\log n > \log \log n$$

$$g(n) \geq f(n)$$

$$f(n) \leq g(n)$$

$$f(n) = O(g(n))$$

④

Important Pattern

##

Ex: $n = 2^4$
value of $\log_2 2^4 = 4 \times \log_2 2$
 $= 4 \times 1$
 $= 4$

$\log_2 n =$ after how many logs
termination condition
or 1 will come

$\log n$

n में $\log$ की कितनी बार
Base condition आएगा

$n = 2^4$
 $\log n = \log 2^4$
 $= 4$

$\log n = 4$
 $\log \log n = \log 4$
 $= 2$

$\log \log n = 2$
 $\log \log \log n = \log 2$
 $= 1$

3 times

$\log n = 4$
 $\log \log n = 3$

Q4



$\log n = 2$
 $\log^* n = 2$

$f(n) = 5$
 $g(n) = 2$

$f(n) \geq$

$f(n) = \log^*(\log n)$
 $g(n) = \log(\log^* n)$

Compare

$\Rightarrow \log 6$

$\log n = \log 2^{2^{2^{2^2}}}$

$\log^* n =$

$\log^* n = 5$

$\log^* (2^{2^{2^{2^2}}}) = 2^{2^{2^{2^2}}}$
 $\log^* 2^{2^{2^{2^2}}} = 2^{2^{2^{2^2}}}$
 $\log^* 2^{2^{2^{2^2}}} = 2^{2^{2^{2^2}}}$

①
 ①

ਪ੍ਰੋਟੋ ਲੋਗ ਟੀ ਵਲ ਵਾਲੂ ਫੀ
and
ਪ੍ਰੋਟੋ ਲੋਗ ਟੀ ਵਲ ਵਾਲੂ
ਕੋਈ ਏਨੀ)

Thanks
DAC



nenu

